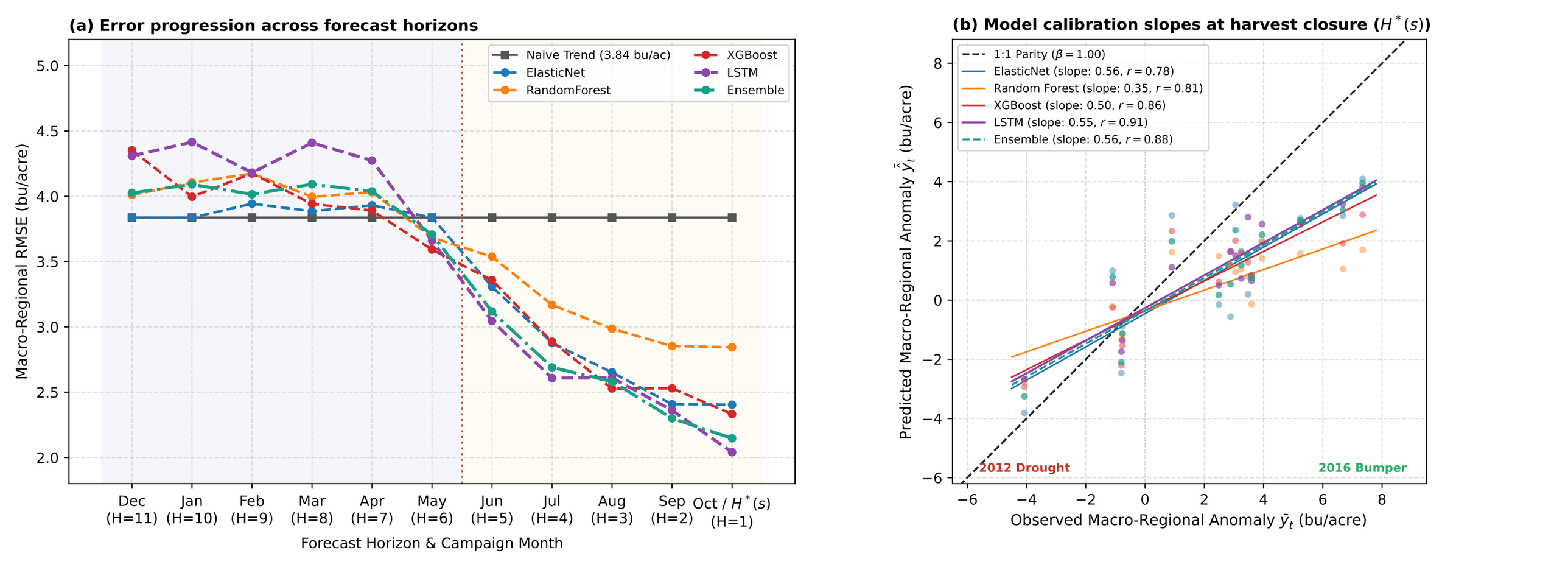
(a) Error progression across forecast horizons
Naive Trend (3.84 bu/ac)
ElasticNet
RandomForest
XGBoost
LSTM
Ensemble
Macro-Regional RMSE (bu/acre)
5.0
4.5
4.0
3.5
3.0
2.5
2.0
Dec (H=11)
Jan (H=10)
Feb (H=9)
Mar (H=8)
Apr (H=7)
May (H=6)
Jun (H=5)
Jul (H=4)
Aug (H=3)
Sep (H=2)
Oct / H*(s) (H=1)
Forecast Horizon & Campaign Month
(b) Model calibration slopes at harvest closure (H*(s))
1:1 Parity (β = 1.00)
ElasticNet (slope: 0.56, r = 0.78)
Random Forest (slope: 0.35, r = 0.81)
XGBoost (slope: 0.50, r = 0.86)
LSTM (slope: 0.55, r = 0.91)
Ensemble (slope: 0.56, r = 0.88)
Predicted Macro-Regional Anomaly ŷ_t (bu/acre)
Observed Macro-Regional Anomaly ȳ_t (bu/acre)
2012 Drought
2016 Bumper
8
6
4
2
0
−2
−4
−6